

☒ HARD

☐ ADVANCED MATH

Advanced Math

10 questions · ~15 min

10

SAT QUESTION BANK

2026-05-29

Q1

GRID-IN ADVANCED MATH

$$y = -2.5$$

$$y = x^2 + 8x + k$$

In the given system of equations, k is a positive integer constant. The system has no real solutions. What is the least possible value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$h(x) = 2(x - 4)^2 - 32$$

The quadratic function h is defined as shown. In the xy -plane, the graph of $y = h(x)$ intersects the x -axis at the points $(0, 0)$ and $(t, 0)$, where t is a constant. What is the value of t ?

- A. 1
- B. 2
- C. 4
- D. 8

Q3

GRID-IN

ADVANCED MATH

$$x - 9 + 45 = 63$$

What is the sum of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$f(x) = (x - 1)(x + 3)(x - 2)$$

In the xy -plane, when the graph of the function f , where $y = f(x)$, is shifted up 6 units, the resulting graph is defined by the function g . If the graph of $y = g(x)$ crosses through the point $4, b$, where b is a constant, what is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

ADVANCED MATH

$$x^2 - 2x - 9 = 0$$

One solution to the given equation can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

- A. 8
- B. 10
- C. 20
- D. 40

Q6

ADVANCED MATH

The first term of a sequence is 9. Each term after the first is 4 times the preceding term. If w represents the n th term of the sequence, which equation gives w in terms of n ?

- A. $w = 49^n$
- B. $w = 49^{n-1}$
- C. $w = 94^n$
- D. $w = 94^{n-1}$

Q7

ADVANCED MATH

In the xy -plane, the graph of the equation $y = -x^2 + 9x - 100$ intersects the line $y = c$ at exactly one point. What is the value of c ?

- A. $-\frac{481}{4}$
- B. -100
- C. $-\frac{319}{4}$
- D. $-\frac{9}{2}$

Q8

ADVANCED MATH

A right rectangular prism has a height of 9 inches. The length of the prism's base is x inches, which is 7 inches more than the width of the prism's base. Which function V gives the volume of the prism, in cubic inches, in terms of the length of the prism's base?

- A. $Vx = xx + 9x + 7$
- B. $Vx = xx + 9x - 7$
- C. $Vx = 9xx + 7$
- D. $Vx = 9xx - 7$

$$\frac{4x^2}{x^2-9} - \frac{2x}{x+3} = \frac{1}{x-3}$$

What value of x satisfies the equation above?

- A. -3
- B. $-\frac{1}{2}$
- C. $\frac{1}{2}$
- D. 3

The population P of a certain city y years after the last census is modeled by the equation below, where r is a constant and

$$P_0$$

is the population when

$$y = 0$$

.

$$P = P_0(1 + r)^y$$

If during this time the population of the city decreases by a fixed percent each year, which of the following must be true?

- A. $r < -1$
- B. $-1 < r < 0$
- C. $0 < r < 1$
- D. $r > 1$